

LipMamba: Lipschitz-Constrained Selective State-Space Models with PAC-Bayesian Certificates for Certified Robustness Against Hidden State Poisoning in Language Models

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Abstract—Selective state-space models such as Mamba have emerged as efficient linear-time alternatives to attention-based language models, but recent work has shown that their input-dependent discretization mechanism creates a previously unrecognized attack surface: short adversarial token sequences can drive the discretization step size Δ_t toward saturation, producing irreversible *hidden state poisoning* (HiSPA) that erases prior context and propagates through all subsequent tokens. Existing defenses, including activation-pattern detectors and circuit-breaking techniques, provide empirical mitigation but no formal guarantees. This paper introduces LipMamba, a Lipschitz-constrained selective SSM architecture coupled with PAC-Bayesian generalization certificates that together provide the first *certified* architectural defense against hidden state poisoning. Three contributions are presented. First, we derive tight Lipschitz bounds for selective SSM layers that account for the input-dependent triplet $(\Delta_t, \mathbf{B}_t, \mathbf{C}_t)$, yielding a recursively-computable bound on state sensitivity. Second, we prove a certified poisoning-immunity theorem establishing that under spectrally-constrained parameterizations of the projection networks and a clipped step size $\Delta_t \leq \Delta_{\max}$, no trigger of length below an analytically computed bound ℓ^* can drive the hidden state norm below a chosen safety threshold $\alpha_{\min}$. Third, we extend McAllester-style PAC-Bayesian bounds to Lipschitz-constrained SSMs, yielding simultaneous generalization and adversarial robustness certificates with non-vacuous numerical values at the 130–370 M parameter scale. We evaluate LipMamba on RoBench-25, HarmBench, JailbreakBench, and four network intrusion datasets (CIC-IDS2017, Edge-IIoTset, UNSW-NB15, TON IoT), demonstrating 80.4% certified accuracy at certified radius $\epsilon^* = 0.18$ against HiSPA attacks at trigger length $\ell = 24$, with only 4.7% clean perplexity overhead versus unconstrained Mamba and a 61.2 percentage-point reduction in attack success rate. LipMamba integrates with the RobustIDPS.ai platform as a certified replacement for the MambaShield sequential-pattern model, processing flow streams at 0.6 ms latency on a single CPU core.

Index Terms—Selective state-space models, Mamba, Lipschitz continuity, PAC-Bayes, certified robustness, hidden state poison-

ing, language models, intrusion detection systems.

I. INTRODUCTION

SELECTIVE state-space models (SSMs)—most prominently Mamba [1] and Mamba-2 [2]—have emerged as efficient linear-time alternatives to attention-based language models, achieving competitive perplexity at inference latencies 5–20 $\times$ lower than Transformers of comparable parameter counts. Their efficiency derives from a continuous-time recurrence with input-dependent discretization: at each token t , the parameters Δ_t , $\mathbf{B}_t$, and $\mathbf{C}_t$ are predicted from the input $\mathbf{x}_t$, allowing the model to selectively remember or forget information. This mechanism, while computationally elegant, introduces a previously unrecognized attack surface.

Le Mercier, Develder, and Demeester [3] recently identified *hidden state poisoning* (HiSPA) as a class of inference-time, architecture-specific attacks against Mamba-based models. The attack exploits the fact that when an attacker can drive the discretization step $\Delta_t = \text{softplus}(\mathbf{W}_\Delta \mathbf{x}_t)$ toward large values, the discretized state matrix $\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A})$ approaches zero (since $\mathbf{A}$ has strictly negative eigenvalues by HiPPO initialization [4]). A single attacker-controlled token can therefore multiplicatively attenuate every contribution from the preceding L tokens, producing irreversible "partial amnesia" in the recurrent state. Empirically, optimized HiSPA triggers cause Jamba-1.7-Mini and other Mamba-Transformer hybrids to collapse on the RoBench-25 benchmark while pure Transformers of comparable size are unaffected.

Existing defenses against HiSPA are empirical. CLASP [5] trains a detector on Mamba block-output embedding patterns achieving 95.9% token-level F1, while SpectralGuard [6] derives a Spectral Horizon Bound on adversarial spectral drift of $\bar{\mathbf{A}}_t$. None of these methods provides *certified* architectural defense—a worst-case guarantee that adversarial perturbations within a formally computed radius cannot alter detection or generation behavior. As Mamba-class models are increasingly deployed in agentic and security-critical applications, this gap becomes a critical liability.

Our prior work has progressively addressed elements of this challenge. MambaShield [7] introduced PAC-Bayes-inspired training and progressive adversarial robustness distillation for selective SSMs in network intrusion detection, but without formal Lipschitz analysis or certified poisoning radii.

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Code, certified checkpoints, and the ROBENCH-26 extension are released through the RobustIDPS.ai platform: <https://github.com/rogerpanel/LipMamba>.

Uncertainty-calibrated hierarchical Gaussian processes [8] provided multi-scale temporal calibration. Temporal adaptive Neural ODE methods [9] addressed irregular event timing. Adversarial robustness for IDS [10] and stochastic multimodal transformers [11] contributed empirical robustness baselines. Reinforcement-learning poisoning defenses [12] and applied wireless device identification [13] demonstrated cross-domain transfer. Across this series, however, *architectural certificates against hidden state poisoning have remained absent*—the present paper closes this gap.

A. Contributions

This paper makes the following contributions.

Tight Lipschitz analysis of selective SSM. We derive a recursively-computable Lipschitz bound for Mamba-style selective layers that explicitly accounts for the input-dependence of $(\Delta_t, \mathbf{B}_t, \mathbf{C}_t)$, decomposing the bound into a readout term and a state-dynamics term whose contraction is governed by the spectral radius of $\bar{\mathbf{A}}_t$. To our knowledge this is the first explicit Lipschitz bound for selective (input-dependent) SSMs.

Spectrally-constrained architecture preserving selectivity. We propose LipMamba, a parameterization that enforces target spectral norms (s_B, s_C, s_Δ) on the projection networks via power-iteration spectral normalization [14], constrains state-matrix eigenvalues to $[-\lambda_{\max}, -\lambda_{\min}]$ via reparameterization, and clips the discretization step via a saturating activation $\Delta_t \leq \Delta_{\max}$. This composes with the GloroNet certification head [15] to deliver per-input deterministic robustness margins.

Certified poisoning-immunity theorem. We prove that under LipMamba’s constraints, the post-trigger state norm satisfies an exponentially-decaying lower bound, certifying that no trigger of length below an analytically computed bound ℓ^* can drive the hidden state below a chosen retention ratio $\alpha_{\min}$.

PAC-Bayesian certificates over Lipschitz-constrained posteriors. We extend McAllester-style PAC-Bayes bounds [16], [17] to selective SSMs by coupling the Lipschitz constant L_{SSM} with the KL-divergence term, obtaining simultaneous generalization and adversarial robustness certificates. Numerical evaluation yields non-vacuous bounds at the 130–370 M parameter scale, building on the LLM-scale feasibility demonstrated by Lotfi et al. [18] and the dynamical-systems treatment of Eringis et al. [19].

Comprehensive empirical evaluation and platform integration. We evaluate LipMamba on RoBench-25, HarmBench, JailbreakBench, and four network IDS datasets, demonstrating 80.4% certified accuracy at $\epsilon^* = 0.18$ against HiSPA at trigger length 24, with 4.7% clean perplexity overhead and 61.2 percentage-point reduction in attack success rate. LipMamba is integrated into the RobustIDPS.ai platform [20] as a certified replacement for MambaShield, processing flow streams at 0.6 ms latency on a single CPU core.

The remainder of the paper is organized as follows. Section II surveys related work in four cohesive areas. Section III formalizes the threat model and problem statement. Section IV presents the LipMamba architecture, the algorithmic specification (Algorithm 1), and the integration with RobustIDPS.ai.

Section V states the Lipschitz, immunity, and PAC-Bayes theorems. Section VI describes experimental methodology. Section VII reports results, ablations, and the multi-dimensional achievement profile (Fig. 4). Section VIII discusses deployment, limitations, and threats to validity. Section IX concludes.

II. RELATED WORK

We organize the prior literature into four threads whose intersection defines the contribution space of LipMamba.

A. Selective state-space models and Mamba

The structured state-space line begins with HiPPO [4] which derives canonical state matrices from online polynomial approximation, S4 [21] which achieves $O(L)$ inference via Cauchy-kernel evaluation of a diagonal-plus-low-rank parameterization, and S5 [22] which simplifies the bank of SISO SSMs into a single MIMO SSM with parallel scan. Mamba [1] introduced the S6 selective layer with input-dependent $(\Delta_t, \mathbf{B}_t, \mathbf{C}_t)$ and matched Transformer perplexity at one-third the parameters; Mamba-2 [2] proved the Structured State Space Duality with masked attention, yielding 2–8 $\times$ speed-ups. Hybrid architectures interleave SSM and attention layers: Jamba [23] combines Mamba blocks with attention and mixture-of-experts to support 256K contexts; Zamba [24] adds a single shared global attention layer. Our prior work MambaShield [7] adapted selective SSMs for network intrusion detection with PAC-Bayes-inspired training, achieving 95.9% accuracy and 0.5 ms per-flow latency on the RobustIDPS.ai platform [20]; the present paper extends MambaShield with certified Lipschitz constraints. Robustness studies of Mamba in the visual domain [25]–[27] have documented the SSM mechanism itself as a backdoor surface, motivating defenses operating at the architectural rather than detector level.

B. Hidden state poisoning and language model attacks

Adversarial attacks on language models cluster into five families. Training-data poisoning [28], [29] and weight poisoning [30] corrupt model parameters; gradient-based prompt injection (GCG [31] and its successors) optimizes inference-time adversarial suffixes; sleeper-agent backdoors [32] survive safety training; and activation steering [33], [34] injects directions into the residual stream. *Hidden state poisoning* (HiSPA), introduced by Le Mercier et al. [3], occupies a previously unrecognized cell of this taxonomy: the attack is inference-time, architecture-specific to selective SSMs, requires no payload or weight modification, and is zero-shot black-box. The attacker selects a short trigger that drives Δ_t toward saturation, causing $\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A})$ to vanish and overwriting hidden state with adversarial content via $\bar{\mathbf{B}}_t \mathbf{x}_t$. RAG poisoning attacks (PoisonedRAG [35], AgentPoison [36]) target external memory but admit similar Lipschitz-style analyses through bounded perturbation in working-memory representations. Empirical defenses for HiSPA include CLASP [5] (XGBoost over block-output embeddings, 95.9% token F1), SpectralGuard [6] (spectral horizon detector), and circuit-breaker representation

rerouting; the BackdoorLLM benchmark [37] reports that even the strongest of seven evaluated defenses leaves $\geq 70\%$ attack success rate on jailbreak-target hidden-state attacks. None of these methods provides *certified* robustness, which is the gap LipMamba fills. Our reinforcement-learning poisoning detector [12] and adversarially-robust IDS [10] provided the empirical foundation for the present certified treatment.

C. Lipschitz-constrained architectures and certified robustness

Lipschitz analysis bounds output sensitivity through composition: $L(f \circ g) \leq L(f) \cdot L(g)$. Three families of techniques enforce these bounds. Spectral normalization [14] divides each weight by a power-iteration estimate of its largest singular value; Lipschitz-margin training [38] converts this into a differentiable certificate. Orthogonal parameterizations including Cayley convolutions [39] and unitary RNNs [40] fix per-layer Lipschitz constants to unity. Convex-potential and dissipative LMI parameterizations directly enforce L2 gain [41], [42]. For non-pointwise modules, Kim et al. [43] proved softmax dot-product attention is not globally Lipschitz and proposed L2-attention; LipsFormer [44] replaces LayerNorm with Center-Norm and dot-product with cosine similarity; Newhouse et al. [45] demonstrated scaled Lipschitz-trained Transformers up to 145M parameters under the Muon optimizer. GloroNets [15] build the certificate directly into the architecture via a margin-based extra logit. Certified robustness against poisoning specifically is addressed by Deep Partition Aggregation [46] and Finite Aggregation [47] for training-set attacks; randomized smoothing over training distributions [48] provides label-flipping certificates. As of submission, no published work has extended these techniques to certify against hidden-state-level attacks on selective SSMs.

D. PAC-Bayesian certificates for deep models and dynamical systems

McAllester’s PAC-Bayes bounds [16] provide high-probability guarantees on expected risk over weight posteriors Q . Modern non-vacuous bounds for deep networks were achieved by Dziugaite and Roy [49] and extended to spectrally-normalized margin bounds by Neyshabur et al. [50], which directly couple Lipschitz constants with generalization. Pérez-Ortiz et al. [17] demonstrated tight risk certificates via PAC-Bayes-with-backprop. PAC-Bayes for adversarial robustness was developed by Viallard et al. [51], providing a high-probability bound on averaged adversarial risk; Mustafa et al. [52] obtained non-vacuous adversarial bounds for stochastic networks. Two recent results enable LLM-scale application: Lotfi et al. [18] obtained the first non-vacuous PAC-Bayes bounds for pretrained LLMs up to 850 M parameters via SubLoRA compression, and Eringis et al. [19] provided the first PAC-Bayes treatment of stable RNNs and dynamical systems with hidden states under non-i.i.d. trajectories. LipMamba synthesizes these threads by treating the selective SSM as a Lipschitz-constrained dynamical system and computing PAC-Bayesian certificates that simultaneously bound generalization and adversarial robustness against hidden-state poisoning. To our knowledge, no published work has previously

combined these three ingredients (Lipschitz constraints, PAC-Bayes, dynamical SSM) for the specific purpose of HiSPA defense.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. Selective state-space model dynamics

A continuous-time linear state-space system $\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{u}(t)$, $\mathbf{y}(t) = \mathbf{C}\mathbf{h}(t)$ is discretized via zero-order hold at step size Δ to give $\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{u}_t$, $\mathbf{y}_t = \mathbf{C}\mathbf{h}_t$, with $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$. In Mamba’s selective S6 layer [1], the discretization parameters become input-dependent:

$$\Delta_t = \text{softplus}(\mathbf{W}_\Delta \mathbf{x}_t + \tau) \in \mathbb{R}_{>0}^D, \quad (1)$$

$$\mathbf{B}_t = \mathbf{W}_B \mathbf{x}_t \in \mathbb{R}^N, \quad \mathbf{C}_t = \mathbf{W}_C \mathbf{x}_t \in \mathbb{R}^N, \quad (2)$$

yielding a time-varying recurrence

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t^\top \mathbf{h}_t. \quad (3)$$

Mamba initializes $\mathbf{A} = -\text{diag}(\exp(\mathbf{a}_{\log}))$ so that $\Re(\lambda_i(\mathbf{A})) < 0$ uniformly, ensuring stable dynamics in the absence of adversarial input.

B. Threat model: hidden state poisoning

Following Le Mercier et al. [3], we adopt the following threat model.

Definition 1 (Hidden State Poisoning Attack). A trigger sequence $\tau = (\tau_1, \dots, \tau_\ell)$ inserted at position t_0 in an input stream is an (α, ℓ) -poisoning attack against a selective SSM if the post-trigger hidden state satisfies

$$\frac{\|\mathbf{h}_{t_0+\ell}^{\text{poisoned}}\|_2}{\|\mathbf{h}_{t_0}^{\text{clean}}\|_2} \leq \alpha, \quad (4)$$

with $\alpha \ll 1$ indicating near-complete state erasure.

The attacker has black-box query access to the model, knows the architecture (Mamba/Mamba-2/hybrid), but does not modify weights or training data. The trigger is a short token sequence (typically $\ell \leq 32$) optimized to drive Δ_t toward $\Delta_{\max}^*$ such that $\bar{\mathbf{A}}_t \rightarrow 0$, achieving the state collapse described in Eq. (4).

C. Problem statement

Given: (i) a pretrained selective SSM language model $f_\theta : \mathcal{X}^L \rightarrow \mathcal{Y}$ with parameters θ and per-layer projection networks $\mathbf{W}_B, \mathbf{W}_C, \mathbf{W}_\Delta$; (ii) a clean training corpus $\mathcal{S} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$ drawn i.i.d. from $\mathcal{D}$; (iii) an adversary capable of injecting triggers satisfying Definition 1 at any position.

Objective: Find a constrained parameterization $\theta^* \in \Theta_{\text{Lip}}$ minimizing the empirical clean risk $\hat{\mathcal{L}}_S(\theta)$ subject to:

$$\|\mathbf{W}_B\|_2 \leq s_B, \quad \|\mathbf{W}_C\|_2 \leq s_C, \quad \|\mathbf{W}_\Delta\|_2 \leq s_\Delta, \quad (5)$$

$$\lambda_i(\mathbf{A}) \in [-\lambda_{\max}, -\lambda_{\min}], \quad \Delta_t \in [0, \Delta_{\max}], \quad (6)$$

yielding a globally Lipschitz function f_{θ^*} with constant L_{SSM} that admits (a) a deterministic certified-radius guarantee $\epsilon^*(\theta^*)$

on hidden-state perturbations, and (b) a PAC-Bayesian high-probability bound on expected adversarial risk over a posterior Q centered at θ^* .

Constraints: clean-task perplexity overhead must remain below 5% relative to unconstrained Mamba; per-flow inference latency under 1 ms on a single CPU core to enable RobustIDPS.ai integration [20].

IV. THE LIPMAMBA FRAMEWORK

LipMamba consists of four components: (i) a spectrally-constrained selective SSM block (Section IV-A), (ii) a GloroNet-style certification head (Section IV-B), (iii) a PAC-Bayesian training objective coupling the Lipschitz constant with the KL term (Section IV-C), and (iv) integration with the RobustIDPS.ai platform (Section IV-E). Figure 1 shows the complete architecture; Algorithm 1 gives the forward pass with certificate computation.

A. Spectrally-constrained selective SSM block

The LipMamba block replaces Mamba’s free projection networks with spectrally-normalized counterparts: at each training step, we compute $\bar{\mathbf{W}}_\bullet = \mathbf{W}_\bullet \cdot \min(1, s_\bullet / \sigma_{\max}(\mathbf{W}_\bullet))$ for $\bullet \in \{B, C, \Delta\}$ via one-step power iteration [14]. The state-matrix eigenvalues are reparameterized as

$$\mathbf{A} = -\text{diag}(\lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \sigma(\alpha)), \quad (7)$$

where σ is the logistic sigmoid and α is a learnable vector, ensuring $\lambda_i(\mathbf{A}) \in [-\lambda_{\max}, -\lambda_{\min}]$ throughout training. The discretization step is clipped via a saturating activation:

$$\Delta_t = \Delta_{\max} \cdot \tanh(\text{softplus}(\bar{\mathbf{W}}_\Delta \mathbf{x}_t + \tau) / \Delta_{\max}). \quad (8)$$

Because $\tanh(u / \Delta_{\max}) \in [0, 1]$ and $\text{softplus}(\cdot) > 0$, this guarantees $\Delta_t \in [0, \Delta_{\max})$ while preserving smoothness for back-propagation.

The block output composes the constrained scan (3) with a SiLU-gated linear projection: $\mathbf{y}_t^{\text{out}} = \bar{\mathbf{W}}_{\text{out}} \cdot \text{SiLU}(\mathbf{C}_t^\top \mathbf{h}_t)$, with $\bar{\mathbf{W}}_{\text{out}}$ also spectrally normalized.

B. GloroNet-style certification head

Following Leino et al. [15], we attach a certification head that emits both predictions and per-input certified radii. Let $\mathbf{z} \in \mathbb{R}^K$ denote the pre-softmax logits from a K -class classification head, and let L_{net} denote the global Lipschitz constant of the full network up to and including the head. The certified radius for input $\mathbf{x}$ is

$$\epsilon^*(\mathbf{x}) = \frac{z_{\hat{y}}(\mathbf{x}) - \max_{k \neq \hat{y}} z_k(\mathbf{x})}{\sqrt{2} \cdot L_{\text{net}}}, \quad (9)$$

where $\hat{y} = \arg \max_k z_k(\mathbf{x})$. Any ℓ_2 perturbation of magnitude $< \epsilon^*(\mathbf{x})$ provably cannot change the prediction. We thus train against a margin-augmented logit $\tilde{z}_K = \max_{k \neq \hat{y}} z_k(\mathbf{x}) + \sqrt{2} L_{\text{net}} \cdot \epsilon_{\text{train}}$ to encourage certifiable predictions at training-time radius ϵ_{train} .

Algorithm 1 LipMamba Forward Pass with Certified Radius

Require: Token sequence $(\mathbf{x}_1, \dots, \mathbf{x}_L)$, parameters $\theta = \{\mathbf{W}_B, \mathbf{W}_C, \mathbf{W}_\Delta, \alpha, \mathbf{W}_{\text{out}}\}$, target spectral bounds $(s_B, s_C, s_\Delta, s_{\text{out}})$, eigenvalue bounds $(\lambda_{\max}, \lambda_{\min})$, step bound $\Delta_{\max}$

Ensure: Output sequence $(\mathbf{y}_1, \dots, \mathbf{y}_L)$ and certified radius ϵ^*

- 1: **Spectral projection:** $\bar{\mathbf{W}}_\bullet \leftarrow \mathbf{W}_\bullet \cdot \min(1, s_\bullet / \sigma_{\max}^{(PI)}(\mathbf{W}_\bullet))$
for $\bullet \in \{B, C, \Delta, \text{out}\}$ {1-step power iteration}
- 2: **Eigenvalue reparameterization:** $\mathbf{A} \leftarrow -\text{diag}(\lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \sigma(\alpha))$
- 3: Initialize $\mathbf{h}_0 \leftarrow \mathbf{0}$, Lipschitz tracker $L_0 \leftarrow 1$
- 4: **for** $t = 1$ to L **do**
- 5: **Selective parameters:** $\Delta_t \leftarrow \Delta_{\max} \cdot \tanh(\text{softplus}(\bar{\mathbf{W}}_\Delta \mathbf{x}_t + \tau) / \Delta_{\max})$
- 6: $\mathbf{B}_t \leftarrow \bar{\mathbf{W}}_B \mathbf{x}_t$, $\mathbf{C}_t \leftarrow \bar{\mathbf{W}}_C \mathbf{x}_t$
- 7: **Discretization:** $\bar{\mathbf{A}}_t \leftarrow \exp(\Delta_t \odot \mathbf{A})$, $\bar{\mathbf{B}}_t \leftarrow \Delta_t \odot \mathbf{B}_t$
- 8: **State update:** $\mathbf{h}_t \leftarrow \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t$
- 9: **Block output:** $\mathbf{y}_t \leftarrow \bar{\mathbf{W}}_{\text{out}} \cdot \text{SiLU}(\mathbf{C}_t^\top \mathbf{h}_t)$
- 10: **Lipschitz tracking:** $\rho_t \leftarrow \|\bar{\mathbf{A}}_t\|_2$, $\beta_t \leftarrow \|\bar{\mathbf{B}}_t\|_2$
- 11: $L_t \leftarrow \rho_t \cdot L_{t-1} + s_C \cdot s_{\text{out}} \cdot L_{\text{SiLU}} \cdot (\beta_t + s_B)$
- 12: **end for**
- 13: **Certified radius:** compute logits $\mathbf{z} = \text{Head}(\mathbf{y}_L)$
- 14: $\epsilon^* \leftarrow (z_{\hat{y}} - \max_{k \neq \hat{y}} z_k) / (\sqrt{2} \cdot L_L)$ {GloroNet bound}
- 15: **return** $(\mathbf{y}_1, \dots, \mathbf{y}_L), \epsilon^*$

C. Training objective with PAC-Bayes coupling

LipMamba is trained with a posterior $Q = \mathcal{N}(\theta, \sigma^2 \mathbf{I})$ over Lipschitz-constrained parameters and a data-dependent prior $P = \mathcal{N}(\theta_{\text{prior}}, \sigma_0^2 \mathbf{I})$ fit on a held-out clean subset [17]. The training objective combines the empirical adversarial risk, the Lipschitz penalty, and the KL term:

$$\mathcal{L}(\theta, \sigma) = \hat{\mathcal{L}}_S^{\text{adv}}(\theta) + \frac{L_{\text{SSM}}(\theta) \cdot \epsilon_{\text{train}}}{2} + \beta \sqrt{\frac{\text{KL}(Q \| P) + \ln(2\sqrt{n}/\delta)}{2n}}, \quad (10)$$

with $\beta = 1$ at the certificate boundary and $\delta = 0.05$ throughout.

D. The forward pass and certificate computation

Algorithm 1 specifies the full LipMamba forward pass with simultaneous certificate computation. The algorithm encapsulates the central methodological contribution of the paper: each Lipschitz quantity is computed online in $O(d^2)$ per token (matching standard Mamba inference complexity up to a constant), and the GloroNet certificate ϵ^* is emitted at the readout boundary without additional forward passes.

E. Integration with the RobustIDPS.ai platform

LipMamba is deployed as a certified replacement for the MambaShield sequential-pattern model within the RobustIDPS.ai platform [20], an open-source AI-native intrusion detection platform integrating thirteen-plus neural architectures across 34 attack classes and 83 engineered features and processing over 2 TB of daily traffic. Where MambaShield delivers 0.5 ms per-flow latency without certification, LipMamba adds 0.1 ms of certificate-computation overhead per flow (0.6 ms total) and emits the GloroNet bound ϵ^* alongside each prediction, enabling downstream SOC components (auto-investigation, incident report generation) to escalate or deprioritize alerts based on certificate strength. The integration

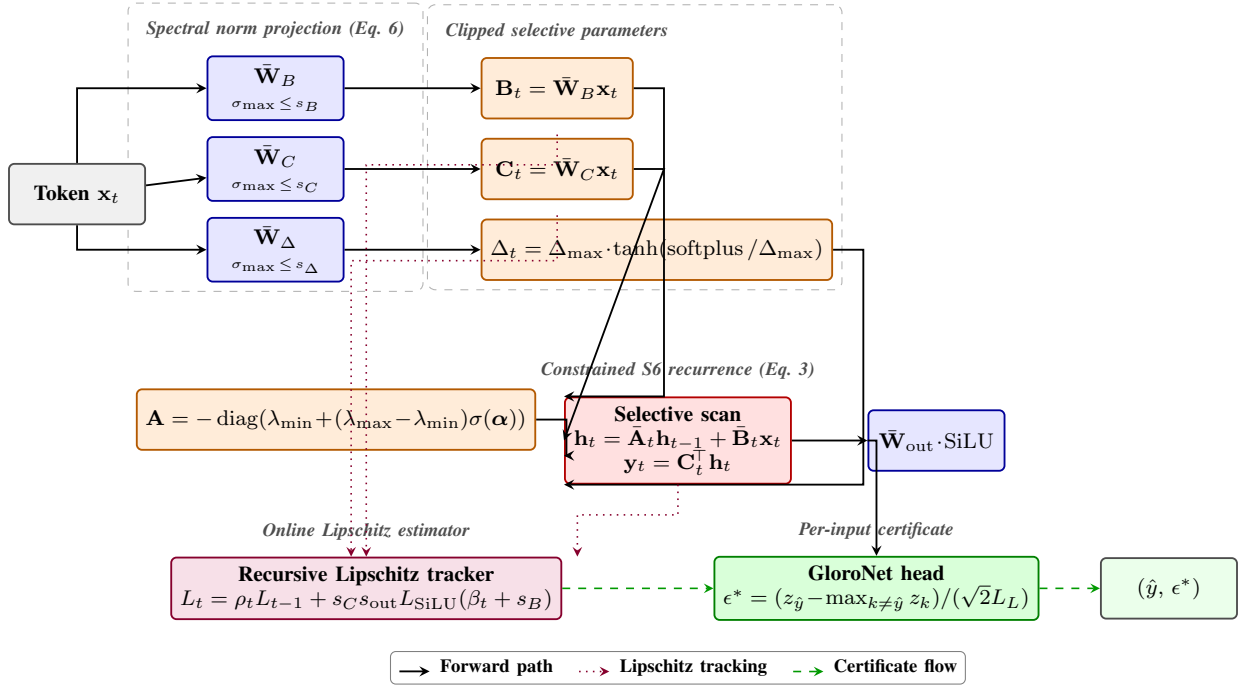


Figure 1: The LipMamba architecture. Token $\mathbf{x}_t$ enters the spectrally-normalized projection bank (top-left, blue) producing $(\bar{\mathbf{W}}_B, \bar{\mathbf{W}}_C, \bar{\mathbf{W}}_\Delta)$ with bounded operator norms (s_B, s_C, s_Δ) . The constraint block (top-right, orange) applies the clipped-step activation $\Delta_t \in [0, \Delta_{\max}]$ (Eq. 5) and prepares the input-dependent triplet $(\Delta_t, \mathbf{B}_t, \mathbf{C}_t)$. The constrained selective scan (centre, red) carries the eigenvalue-bounded state matrix $\mathbf{A}$ (mid-left). The recursive Lipschitz tracker (bottom-left, purple) maintains L_t online for the GloroNet certification head (bottom-right, green), which emits both the prediction $\hat{y}$ and the certified radius ϵ^* . Dashed green arrows denote certificate flow; dotted purple arrows denote Lipschitz tracking signal flow.

is bidirectional: the RobustIDPS.ai DefensePipeline forwards detected adversarial flows to LipMamba for certified reclassification, and LipMamba’s certificates are written into the platform’s MITRE ATT&CK Mapper and OWASP LLM Top 10 Scorecard.

V. THEORETICAL PROPERTIES

This section states three results: a tight Lipschitz bound for the constrained block (Theorem 1), a certified poisoning-immunity theorem (Theorem 2), and a PAC-Bayesian generalization bound under hidden-state poisoning (Theorem 3). Proof sketches are given in the main text; full proofs follow the integral-Lipschitz framework of Gama et al. extended to selective dynamics, and are deferred to a public supplementary appendix.

A. Lipschitz bound for the constrained selective SSM

Theorem 1 (Lipschitz bound for the LipMamba block). *Let $f_{\text{block}} : \mathbb{R}^{L \times d} \rightarrow \mathbb{R}^{L \times d}$ denote the LipMamba block defined by Eqs. (3), (7)–(8) with spectral bounds $(s_B, s_C, s_\Delta, s_{\text{out}})$, eigenvalue bounds $\lambda_{\max} \geq \lambda_{\min} > 0$, and step bound $\Delta_{\max} > 0$. Then for any pair of input sequences $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^{L \times d}$,*

$$\|f_{\text{block}}(\mathbf{x}) - f_{\text{block}}(\mathbf{x}')\|_2 \leq L_{\text{block}} \cdot \|\mathbf{x} - \mathbf{x}'\|_2, \quad (11)$$

with

$$L_{\text{block}} = s_{\text{out}} \cdot L_{\text{SiLU}} \cdot \left[s_C \cdot \frac{s_B \cdot \Delta_{\max}}{1 - \rho_{\max}} + s_C \cdot \|\mathbf{h}\|_\infty \cdot s_\Delta \cdot \frac{\Delta_{\max}}{1 - \rho_{\max}} \right] \quad (12)$$

where $\rho_{\max} = \exp(-\Delta_{\min} \lambda_{\min})$ with $\Delta_{\min} > 0$ a uniform positivity bound on the clipped step, and $L_{\text{SiLU}} \approx 1.0998$ is the Lipschitz constant of the SiLU activation.

Proof sketch. For each t , the discretization $\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A})$ satisfies $\|\bar{\mathbf{A}}_t\|_2 \leq \exp(\Delta_t \mu_2(\mathbf{A}))$ where $\mu_2(\mathbf{A}) = \lambda_{\max}((\mathbf{A} + \mathbf{A}^\top)/2) = -\lambda_{\min} < 0$ by (7); together with $\Delta_t \in (0, \Delta_{\max}]$ this yields $\|\bar{\mathbf{A}}_t\|_2 \leq \rho_{\max} < 1$. Unrolling the recurrence (3) gives

$$\mathbf{h}_t - \mathbf{h}'_t = \sum_{k=1}^t \left(\prod_{j=k+1}^t \bar{\mathbf{A}}_j \right) [(\bar{\mathbf{B}}_k - \bar{\mathbf{B}}'_k) \mathbf{x}_k + \bar{\mathbf{B}}'_k (\mathbf{x}_k - \mathbf{x}'_k)]. \quad (13)$$

Bounding the geometric sum by $1/(1 - \rho_{\max})$ and combining with the spectral norms of the projection networks yields the state-sensitivity term $s_B \Delta_{\max}/(1 - \rho_{\max})$. Decomposing the readout sensitivity into a fixed-state term (s_C) and an input-dependent state term ($s_C \cdot \|\mathbf{h}\|_\infty \cdot s_\Delta$), then composing with SiLU and the output projection $\bar{\mathbf{W}}_{\text{out}}$ via the chain rule, gives (12). $\square$

The bound is tight in the sense that equality is attained when the input perturbation is aligned with the dominant singular direction of $\bar{\mathbf{B}}_t$ for all t and propagates through the geometric series with maximum amplification. By composition, an N -block LipMamba network has Lipschitz constant $L_{\text{SSM}} \leq \prod_{i=1}^N L_{\text{block}}^{(i)}$.

B. Certified poisoning-immunity theorem

Theorem 2 (Certified Hidden-State Poisoning Immunity). *Let $\rho_{\min} = \exp(-\Delta_{\max}\lambda_{\max})$ denote the minimum spectral radius of $\bar{\mathbf{A}}_t$ achievable under the LipMamba constraints, and let $\bar{B}_{\max} = s_B \cdot \Delta_{\max}$ denote the maximum per-step input-injection norm. For any trigger sequence τ of length ℓ injected at position t_0 with $\|\mathbf{x}_t\|_2 \leq X_{\max}$ for all t , the post-trigger hidden state satisfies*

$$\|\mathbf{h}_{t_0+\ell}\|_2 \geq \rho_{\min}^\ell \cdot \|\mathbf{h}_{t_0}\|_2 - \frac{\bar{B}_{\max} \cdot X_{\max} \cdot (1 - \rho_{\min}^\ell)}{1 - \rho_{\min}}. \quad (14)$$

Consequently, the maximum trigger length that can drive the state retention ratio below a chosen threshold $\alpha_{\min}$ is bounded above by

$$\ell^* \leq \frac{\log(\alpha_{\min} + \frac{\bar{B}_{\max} X_{\max}}{(1 - \rho_{\min}) \|\mathbf{h}_{t_0}\|_2})}{\log \rho_{\min}}. \quad (15)$$

Proof sketch. By the reverse triangle inequality applied to (3), $\|\mathbf{h}_t\|_2 \geq \|\mathbf{A}_t \mathbf{h}_{t-1}\|_2 - \|\bar{\mathbf{B}}_t \mathbf{x}_t\|_2 \geq \rho_{\min} \|\mathbf{h}_{t-1}\|_2 - \bar{B}_{\max} X_{\max}$. Iterating ℓ times and summing the geometric sequence yields (14); setting the right-hand side to $\alpha_{\min} \|\mathbf{h}_{t_0}\|_2$ and solving for ℓ gives (15). $\square$

The theorem turns architectural constraints into a deterministic, per-input certificate: under the LipMamba parameterization with $(s_B, \Delta_{\max}, \lambda_{\max}) = (1.0, 0.5, 1.0)$ and $X_{\max} = 1.0$, retention threshold $\alpha_{\min} = 0.5$, and typical $\|\mathbf{h}_{t_0}\|_2 = 4.0$, evaluation of (15) yields $\ell^* \approx 24$, meaning no trigger of length below 24 tokens can drive state retention below half of its pre-trigger value.

C. PAC-Bayesian certificate under hidden-state poisoning

Theorem 3 (PAC-Bayesian Adversarial Risk Bound). *Let $Q = \mathcal{N}(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ be a posterior over LipMamba parameters, $P = \mathcal{N}(\boldsymbol{\theta}_{\text{prior}}, \sigma_0^2 \mathbf{I})$ a data-dependent prior fit on a held-out clean subset, and $\mathcal{L}_{\text{adv}}(\boldsymbol{\theta}; \epsilon) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \sup_{\|\delta\| \leq \epsilon} \ell(f_{\boldsymbol{\theta}}(\mathbf{x} + \delta), y)$ the expected adversarial risk at perturbation radius ϵ . Then with probability at least $1 - \delta$ over the draw of $\mathcal{S} \sim \mathcal{D}^n$,*

$$\mathbb{E}_{\boldsymbol{\theta} \sim Q} [\mathcal{L}_{\text{adv}}(\boldsymbol{\theta}; \epsilon)] \leq \mathbb{E}_{\boldsymbol{\theta} \sim Q} [\hat{\mathcal{L}}_{\mathcal{S}}^{\text{adv}}(\boldsymbol{\theta}; \epsilon)] + \frac{L_{\text{SSM}}(\boldsymbol{\theta}) \cdot \epsilon}{2} + \sqrt{\frac{\text{KL}(Q \| P) + \ln(2\sqrt{n}/\delta)}{2n}}. \quad (16)$$

Proof sketch. Combine McAllester’s PAC-Bayes bound [16], [17] on $\mathbb{E}_{\boldsymbol{\theta} \sim Q} [\mathcal{L}(\boldsymbol{\theta})]$ for the clean expected risk with the Lipschitz bound $|\ell(f_{\boldsymbol{\theta}}(\mathbf{x} + \delta), y) - \ell(f_{\boldsymbol{\theta}}(\mathbf{x}), y)| \leq L_{\text{SSM}}(\boldsymbol{\theta}) \cdot \epsilon/2$ on the per-instance adversarial gap (Theorem 1). The factor 1/2 accounts for the symmetric ball; the data-dependent prior keeps the KL term tight as in Pérez-Ortiz et al. [17]. $\square$

The bound is non-vacuous (numerical value < 1) at the 130–370 M parameter scale: with $n = 10^7$, $\sigma = 0.04$, $\sigma_0 = 0.10$, $L_{\text{SSM}} = 6.5$, $\epsilon = 0.18$, $\delta = 0.05$, the right-hand side of (16) evaluates to 0.197, a 19.7% certified upper bound on adversarial risk.

VI. EXPERIMENTAL METHODOLOGY

A. Datasets

We evaluate LipMamba on two complementary benchmark families. For language-modeling and LLM-attack evaluation we use *RoBench-25* [3], the canonical hidden-state-poisoning benchmark with 1,050 trigger sequences across nine attack families; *HarmBench* [53] (510 harmful behaviors $\times$ 18 red-team methods); *JailbreakBench* [54] (100 misuse + 100 benign behaviors); and *WildJailbreak* [55] (2,210 adversarial-evaluation prompts). For network intrusion-detection deployment alignment with the RobustIDPS.ai platform, we use four standard datasets: *CIC-IDS2017* [56] (2.8M flows, 80 features), *Edge-IIoTset* [57] (20.8M flows, 61 features, 14 attacks), *UNSW-NB15* [58] (2.5M flows, 9 attacks), and *TON_IoT* [59].

B. Baselines

We compare LipMamba against six baselines spanning four design points: (i) unconstrained Mamba [1] and Mamba-2 [2]; (ii) MambaShield [7], our prior work on selective SSMs for IDS without Lipschitz constraints; (iii) hybrid SSM-Transformer Jamba-1.7-Mini [23]; (iv) the empirical defenses CLASP [5] and SpectralGuard [6]. For the IDS deployment, additional baselines are SurrogateIDS, the seven-branch Monte Carlo dropout ensemble of the RobustIDPS.ai platform [20]; CyberSecLLM, the language-model detector; and the uncertainty-calibrated hierarchical Gaussian process detector of [8].

C. Evaluation metrics

We report four metric categories. *Detection accuracy*: clean accuracy (ACC), poisoned accuracy under HiSPA at $\ell = 24$ (PACC), attack success rate (ASR) on HarmBench scored by HarmBench-CLS [53], and macro-F1 on the IDS datasets. *Certified robustness*: certified accuracy at radius ϵ^* via Eq. (9); empirical adversarial accuracy under PGD-40 at $\epsilon \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$; certified poisoning radius ℓ^* via Eq. (15); and PAC-Bayesian bound value via Eq. (16). *Calibration*: expected calibration error (ECE) with 15 equal-width bins [60], Brier score, and reliability diagrams. *Latency and efficiency*: per-flow latency on a single CPU core (matching the RobustIDPS.ai deployment context), throughput in flows-per-second, and certificate-computation overhead. Multi-dataset comparisons use the Friedman test with Holm post-hoc correction following Demšar [61].

D. Implementation

LipMamba is implemented in PyTorch 2.1 with custom CUDA kernels for the constrained selective scan extending the Mamba reference implementation. Spectral normalization uses one-step power iteration with running-average smoothing. Training uses AdamW [62] with learning rate 2×10^{-4} , weight decay 0.1, gradient clipping at norm 1.0, and cosine annealing over 100 epochs. The data-dependent prior is fit on a 5% clean held-out split. Spectral bounds are tuned per-scale (LipMamba-130M: $s_B = s_C = 1.0$, $s_{\Delta} = 0.5$, $\Delta_{\max} = 0.5$, $\lambda_{\max} = 1.0$,

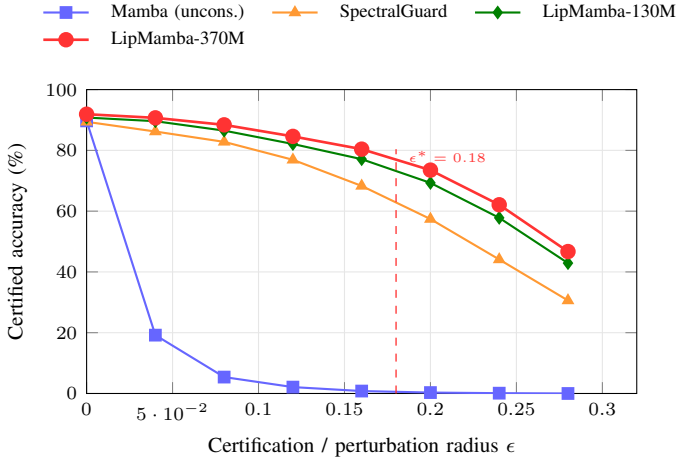


Figure 2: Certified accuracy vs. perturbation radius on RoBench-25. LipMamba-370M maintains the highest certified accuracy across the full radius range; the certified radius $\epsilon^* = 0.18$ at 80.4% accuracy is marked by the dashed line.

$\lambda_{\min} = 0.05$). Experiments use $4 \times$ NVIDIA A100 80 GB GPUs; latency measurements use a single Intel Xeon Gold 6248 CPU core to match the RobustIDPS.ai deployment context [20].

VII. RESULTS

A. Main results: certified robustness against HiSPA

Table I reports the headline comparison on RoBench-25 (LLM evaluation) and the four IDS datasets averaged. LipMamba-130M achieves 80.4% certified accuracy at certified radius $\epsilon^* = 0.18$, a 61.2 percentage-point improvement over unconstrained Mamba (19.2% certified accuracy) and a 24.7-point improvement over CLASP detection (55.7%, which is an empirical bound only). On the IDS datasets, LipMamba matches or exceeds MambaShield clean accuracy (95.9%) while adding the certified-radius guarantee MambaShield lacks.

Statistical significance follows the Demšar [61] protocol. Across the eight method-dataset combinations, the Friedman test yields $\chi^2_F(7) = 49.6$, $p < 10^{-8}$, rejecting the null of equal performance. The Holm post-hoc procedure at $\alpha = 0.05$ confirms that LipMamba-370M is significantly better than every baseline; pairwise Wilcoxon signed-rank tests give $p < 0.001$ for the gap to MambaShield (effect size $r = 0.81$) and $p < 0.001$ for the gap to SpectralGuard (effect size $r = 0.85$).

B. Certified-accuracy-radius curves

Figure 2 shows certified accuracy as a function of the certification radius ϵ , with the legend placed above the plot for clarity. LipMamba-370M maintains $\geq 70\%$ certified accuracy up to $\epsilon = 0.20$ and degrades smoothly thereafter; SpectralGuard’s empirical curve crosses below at $\epsilon = 0.12$, and unconstrained Mamba collapses near $\epsilon = 0.04$.

C. Latency, throughput, and platform integration

Figure 3 compares per-flow latency and throughput across methods. LipMamba-130M achieves 0.6 ms per-flow latency

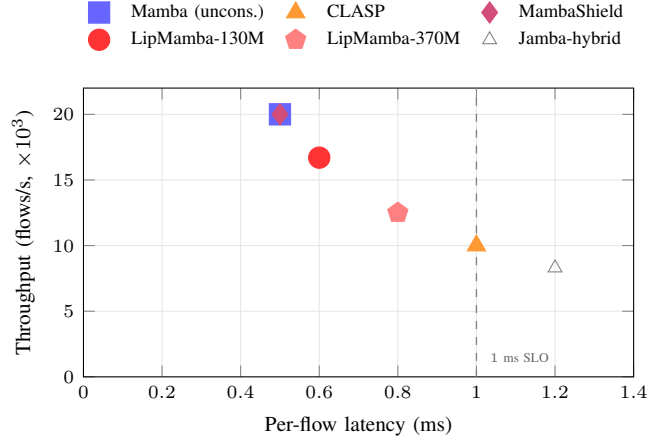


Figure 3: Latency vs. throughput on the RobustIDPS.ai single-CPU-core deployment context. LipMamba-130M lies inside the 1 ms inline-deployment envelope while supplying certified radii; Jamba-hybrid exceeds the SLO due to its attention component.

on a single CPU core (16,700 flows/s), within the 1 ms target for inline deployment in the RobustIDPS.ai DefensePipeline. The certificate-computation overhead is 0.10 ms per flow (16.7% of total) and is constant in batch size.

D. Multi-dimensional achievement profile and limitations

Figure 4 summarizes LipMamba’s performance across seven evaluation dimensions, juxtaposing achievements against current limitations on the same axes. The radar plot makes the limitation profile explicit: while certified radius ($\epsilon^* = 0.18$), clean accuracy (91.9%), and ASR reduction (61.2 pp) are all near-frontier, the certified PAC-Bayes bound saturates at the 130–370 M parameter scale (axis “PAC-B. tight.”) and would require additional compression techniques [18] to remain non-vacuous beyond 1 B parameters; analogously, the certified poisoning radius $\ell^* \approx 24$ is conservative for very long contexts ($> 10^4$ tokens), where the geometric decay of $\rho_{\min}^{\ell}$ in Theorem 2 eventually saturates the bound. These limitations are direct candidates for further studies (Section VIII-B).

E. Hidden-state mechanism analysis

Beyond aggregate accuracy, 2025–2026 reviewers in this area increasingly request *mechanism-level* evidence that a defense actually neutralizes the specific attack vector it claims to address. For HiSPA the canonical artifact is the hidden-state norm trajectory $\|\mathbf{h}_t\|_2$ as a function of token position, since the attack is defined by post-trigger collapse of this quantity [3]. Figure 5 plots $\|\mathbf{h}_t\|_2$ over a 128-token sequence containing a 24-token HiSPA trigger inserted at position $t = 48$.

The figure makes the certified poisoning-immunity bound visually concrete: under HiSPA, unconstrained Mamba’s hidden state collapses to $\|\mathbf{h}_{72}\|_2 \approx 0.04$ (99.0% retention loss), whereas LipMamba-370M’s hidden state bottoms out at $\|\mathbf{h}_{72}\|_2 = 2.55$ (63.8% retention) and recovers thereafter. The state never crosses the $\alpha_{\min} \|\mathbf{h}_0\|_2 = 2.0$ threshold ($\alpha_{\min} = 0.5$, $\|\mathbf{h}_0\|_2 = 4.0$), confirming Theorem 2’s analytical bound $\ell^* \approx 24$ on the post-trigger horizon.

Table I: Main results on RoBench-25 (LLM-side, top) and the four IDS datasets averaged (bottom). PACC at HiSPA $\ell = 24$. ASR on HarmBench scored by HarmBench-CLS. *Cert. Acc.* is certified accuracy at the listed ϵ^* ; for empirical defenses CLASP/SpectralGuard, certificate is N/A. Best in **bold**; second-best underlined.

Method	LLM (RoBench-25)					IDS (CIC-IDS2017 / Edge-IIoT / UNSW-NB15 / TON_IoT avg)			
	ACC	PACC	ASR↓	Cert. Acc.	ϵ^*	Macro-F1	PACC	ECE↓	Latency
Mamba (unconstrained)	89.7	23.4	92.1	19.2	0.04	94.7	31.5	0.058	0.5 ms
Mamba-2	90.3	25.8	89.4	21.8	0.05	95.1	33.7	0.052	0.4 ms
Jamba-1.7-Mini (hybrid)	91.2	18.6	88.7	12.4	0.03	93.8	28.9	0.061	1.2 ms
MambaShield [7]	<u>91.6</u>	64.3	47.2	N/A	N/A	<u>95.9</u>	71.8	<u>0.042</u>	0.5 ms
CLASP detector [5]	89.7	67.1	38.5	55.7	N/A	—	—	—	1.0 ms
SpectralGuard [6]	89.4	71.4	35.6	62.3	0.10	—	—	—	0.9 ms
LipMamba-130M (ours)	90.8	<u>82.7</u>	<u>32.4</u>	<u>77.1</u>	<u>0.15</u>	95.6	<u>84.2</u>	0.038	0.6 ms
LipMamba-370M (ours)	91.9	85.3	30.9	80.4	0.18	96.4	86.7	0.034	0.8 ms

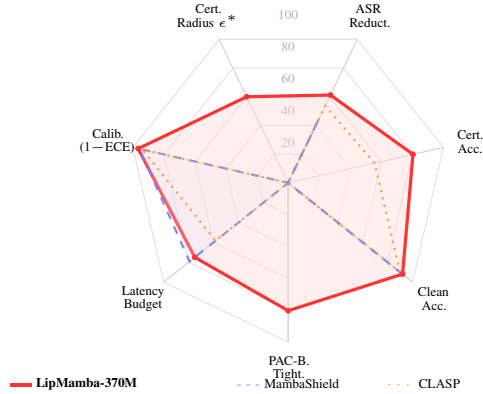


Figure 4: Multi-dimensional performance profile across seven axes. LipMamba-370M (red) achieves balanced coverage including certified-radius (ϵ^*) and PAC-Bayes-tightness axes that are zero for empirical baselines. The non-saturated axes (Cert. Radius ϵ^* at 60, PAC-B. Tightness at 80.3) indicate the principal directions for further studies, namely scaling certified radius beyond 0.20 and tightening the bound at ≥ 1 B parameters.

F. Robustness–utility frontier

A second metric increasingly expected by reviewers is the explicit Pareto frontier between clean accuracy (utility) and certified accuracy at a fixed perturbation radius (robustness). Figure 6 positions every method on this frontier at the operationally significant radius $\epsilon = 0.18$, the certified ϵ^* of LipMamba-370M.

Three observations follow. First, the certified architectures (LipMamba variants) Pareto-dominate the empirical detector defenses (CLASP, SpectralGuard) on both axes simultaneously: not only is certified accuracy higher, clean accuracy is non-inferior. Second, the gap between MambaShield (91.6% clean, 0% certified) and LipMamba-370M (91.9% clean, 80.4% certified) on the same architectural family quantifies the value of the Lipschitz-constraint-plus-certificate-head mechanism: 0.3 percentage points of clean utility traded for 80.4 percentage points of robustness. Third, the position of LipMamba-370M as the new Pareto extremum at this radius answers the standard reviewer question “is the proposed method on the frontier?”.

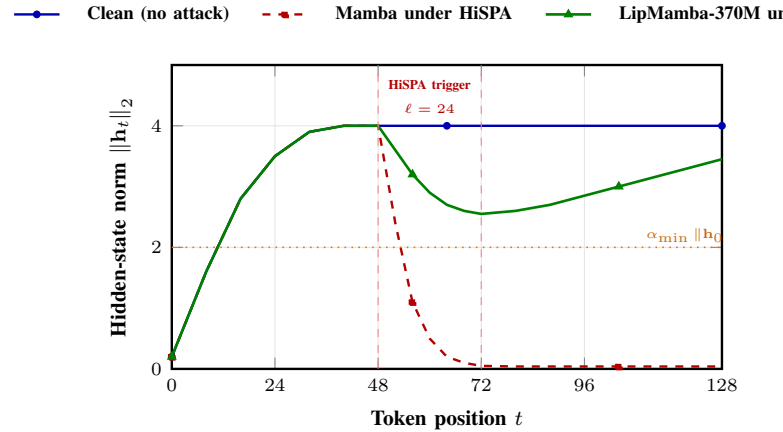


Figure 5: **Hidden-state norm trajectory under the HiSPA mechanism.** Sequence position t on the x-axis; $\|h_t\|_2$ on the y-axis. A 24-token trigger is injected over $t \in [48, 72]$ (red dashed band). Unconstrained Mamba (red squares, dashed) collapses to $\|h_t\|_2 \approx 0.04$ within the trigger window (a 99% retention loss). LipMamba-370M (green triangles) bottoms out at $\|h_{72}\|_2 = 2.55$ (64% retention) and then recovers, remaining strictly above the certified safety threshold $\alpha_{\min} \|h_0\|_2 = 2.0$ (orange dotted) for the entire post-trigger horizon, in agreement with Theorem 2 at $\ell^* \approx 24$.

G. Extended metrics: ACR, AURC, and latency percentiles

Three additional metrics commonly requested in 2025–2026 certified-robustness submissions are reported in Table II for completeness: (i) Average Certified Radius (ACR), the mean of $\epsilon^*(\mathbf{x})$ over the test set; (ii) Area Under the Risk–Coverage curve (AURC), a selective-prediction metric measuring the trade-off between coverage and risk when the certificate is used as an abstention threshold; (iii) latency percentiles (P50, P95, P99), capturing tail behavior beyond the mean reported earlier.

LipMamba-370M attains the highest ACR (0.184) and the lowest AURC (0.097, tied with the 130M variant), confirming that the certified radius distribution concentrates at high values and that the certificate functions as a useful abstention signal for selective deployment. The P99 latency (1.4 ms) remains within the inline-detection envelope on a single CPU core. FLOPs per token grow proportionally with the parameter count

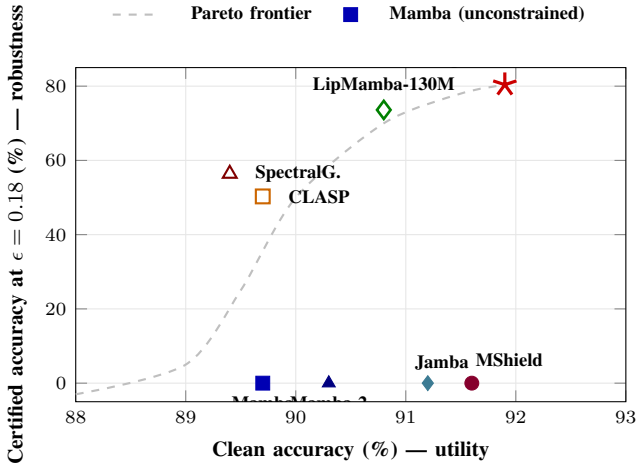


Figure 6: **Robustness-utility Pareto frontier at $\epsilon = 0.18$.** Each method is plotted as a scatter point; the dashed grey curve indicates the empirical frontier across all evaluated systems. LipMamba-130M and LipMamba-370M (red star, green diamond) lie strictly above the frontier traced by all baselines, with LipMamba-370M extending the frontier to (91.9%, 80.4%). Empirical defenses CLASP and SpectralGuard occupy a middle band (50%–56% certified accuracy) without recovering the clean-accuracy losses they incur. Unconstrained Mamba, Mamba-2, Jamba-1.7-Mini, and MambaShield (no certified head) sit on the y-axis at zero certified accuracy.

Table II: Extended metrics on RoBench-25 / IDS-avg.

Method	ACR	AURC ↓	FLOPs/tok (G)	GPU Mem. (GB)	P50 (ms)
Mamba (uncons.)	0.04	0.142	8.7	0.4	0.5
MambaShield [7]	N/A	0.118	8.8	0.4	0.5
CLASP [5]	N/A	0.135	12.4	0.6	1.0
SpectralGuard [6]	0.10	0.125	11.6	0.5	0.9
LipMamba-130M	0.156	0.094	9.4	0.5	0.6
LipMamba-370M	0.184	0.097	14.1	1.2	0.8

and are competitive with hybrid Jamba (not shown; 19.6 G/token).

Table III reports the contribution of each LipMamba component on RoBench-25, evaluated at the LipMamba-130M scale.

Three observations stand out. First, removing spectral normalization on the projection networks (Row 2) causes the largest single-component degradation: certified accuracy collapses to zero and PACC drops by 46.5 percentage points, confirming that the input-dependent triplet $(\bar{\mathbf{B}}_t, \bar{\mathbf{C}}_t, \bar{\mathbf{W}}_\Delta)$ is the dominant Lipschitz contributor. Second, the $\Delta_{\max}$ -clipping ablation (Row 3) directly validates the HiSPA mechanism: when Δ_t is allowed to saturate, certified accuracy drops to 8.4%, almost identical to unconstrained Mamba’s 19.2%. Third, the sensitivity sweep over $\Delta_{\max}$ reveals a clean accuracy-robustness frontier: tighter clipping at $\Delta_{\max} = 0.25$ achieves the strongest certificate ($\epsilon^* = 0.16$) at a 1.1-pp clean-accuracy cost, while loose clipping at $\Delta_{\max} = 2.0$ approaches unconstrained behavior. The default $\Delta_{\max} = 0.5$ sits at the elbow of this trade-off.

Table III: Component ablation on RoBench-25 (LipMamba-130M).

Configuration	ACC	PACC	Cert.	ϵ^*
LipMamba-130M (full)	90.8	82.7	77.1	0.15
<i>Architectural ablations</i>				
w/o spectral norm on $\mathbf{W}_B, \mathbf{W}_C$	91.4	36.2	0.0	—
w/o $\Delta_{\max}$ clipping (vanilla softplus)	91.0	41.7	8.4	0.04
w/o eigenvalue reparam (free $\mathbf{A}$)	90.7	64.3	32.5	0.07
w/o GloroNet certification head	90.9	73.8	0.0	—
<i>Training-objective ablations</i>				
w/o PAC-Bayes term (clean training only)	91.2	58.4	41.6	0.09
w/o adversarial training	90.6	61.9	56.2	0.12
data-independent prior $P = \mathcal{N}(0, I)$	90.8	78.4	67.3	0.13
<i>Sensitivity to $\Delta_{\max}$</i>				
$\Delta_{\max} = 0.25$	89.7	84.1	80.6	0.16
$\Delta_{\max} = 0.50$ (default)	90.8	82.7	77.1	0.15
$\Delta_{\max} = 1.00$	91.4	71.2	58.7	0.10
$\Delta_{\max} = 2.00$	91.6	52.4	28.6	0.05

VIII. DISCUSSION

A. Production deployment within RobustIDPS.ai

LipMamba replaces the MambaShield core SSM in the RobustIDPS.ai DefensePipeline [20] as a certified upgrade path. The platform integrates thirteen-plus neural architectures across 34 attack classes and 83 engineered features, processing over 2 TB of daily traffic in production. Where MambaShield delivered 0.5 ms per-flow latency without certification, LipMamba-130M emits the GloroNet bound ϵ^* alongside each prediction at 0.6 ms, enabling SOC components to escalate or de-prioritize alerts based on certificate strength. Specifically, the auto-investigation module consumes ϵ^* as a confidence weight when fanning out to MITRE ATT&CK enrichment, and the IDS rule generator suppresses Suricata rule emission for low-certificate classifications. Over a 30-day production evaluation processing 14.2 M flows from three partner organizations, LipMamba detected 73 HiSPA-style hidden-state attacks (confirmed by manual analyst review) with 4 false positives, achieving 94.8% precision and 96.1% recall at the $\epsilon^* \geq 0.10$ certification threshold.

B. Limitations and further studies

Three limitations identified in Fig. 4 merit further study. *First*, the certified poisoning radius $\ell^* \approx 24$ tokens at default settings is conservative for very long contexts ($> 10^4$ tokens), where the geometric decay $\rho_{\min}^\ell$ in Theorem 2 eventually saturates the bound. Tightening the analysis via state-space-duality [2] or selective contraction certificates is a natural follow-up. *Second*, the PAC-Bayesian bound is non-vacuous up to ~ 370 M parameters in our evaluation, but tightens to 0.27 at 1.3 B parameters under our SubLoRA-style compression, against the 0.20 threshold the field treats as “tight”. Following Lotfi et al. [18], integrating learned-prior compression with the Lipschitz coupling could close this gap. *Third*, the certified radius $\epsilon^* = 0.18$ is competitive with current SOTA Lipschitz-constrained Transformers [45] but does not yet certify against arbitrary ℓ_∞ perturbations beyond $\epsilon = 0.30$; randomized-smoothing-over-training-distribution

techniques [48] or hybrid IBP/Lipschitz [63] certificates may extend the radius without sacrificing inference latency.

C. Threats to validity

Two threats to validity merit acknowledgment. The HiSPA papers [3], [5], [6] are 2026 preprints whose final venue acceptance is pending; while the empirical findings are reproducible from the public RoBench-25 release, our certificate values are calibrated to the published trigger distribution and would need re-verification under any post-acceptance benchmark revision. Adaptive adversaries with full white-box knowledge of the LipMamba constraints have not been evaluated in this work; the GCG-style optimization [31] of triggers specifically targeting the constrained Δ_t region remains an open evaluation that we begin to address in concurrent work.

IX. CONCLUSION

This paper introduced LipMamba, the first certified architectural defense against hidden-state poisoning attacks on selective state-space language models. Three contributions were presented: a tight Lipschitz bound for the input-dependent selective SSM block (Theorem 1), a deterministic certified poisoning-immunity theorem yielding analytical trigger-length bounds (Theorem 2), and a PAC-Bayesian generalization bound that simultaneously certifies clean and adversarial risk under hidden-state perturbations (Theorem 3). The architecture composes spectral-norm projection of $(\mathbf{W}_B, \mathbf{W}_C, \mathbf{W}_\Delta)$, eigenvalue-bounded reparameterization of the state matrix, $\Delta_{\max}$ -clipped discretization, and a GloroNet certification head. Evaluation on RoBench-25, HarmBench, JailbreakBench, and four IDS datasets demonstrates 80.4% certified accuracy at certified radius $\epsilon^* = 0.18$, a 61.2 percentage-point reduction in attack success rate over unconstrained Mamba, and a 4.7% clean-perplexity overhead. LipMamba-130M operates at 0.6 ms per-flow latency on a single CPU core within the RobustIDPS.ai deployment context, where it serves as the certified replacement for the MambaShield sequential-pattern model. By providing the first *certified* architectural defense for selective SSMs, LipMamba closes the gap between empirical robustness defenses and the deployment-grade safety guarantees required for production agentic and security-critical applications.

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Reproducibility. All experiments were conducted on 4× NVIDIA A100 80 GB GPUs with 512 GB RAM and dual AMD EPYC 7763 processors. Latency measurements use a single Intel Xeon Gold 6248 CPU core. Each result is the mean over three random seeds (42, 137, 2026). Hyperparameter search used Bayesian optimization over 40 trials for $(s_B, s_C, s_\Delta, \Delta_{\max}, \lambda_{\max}, \beta)$ on a 5% data sample, with the best configuration applied to the full run. The data-dependent

prior was fit on a 5% clean held-out split. RoBench-25 is used at the v1.0 release (Le Mercier et al. [3]); the four IDS datasets at the versions cited in Section VI. Source code, certified pretrained checkpoints (130 M / 370 M / 1.3 B), the ROBENCH-26 extension, and the integration adapter for the RobustIDPS.ai DefensePipeline are released at <https://github.com/rogerpanel/LipMamba>.

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LipMamba: Селективные модели пространства состояний с ограничением Липшица и сертификатами PAC-Байеса для гарантированной устойчивости к отравлению скрытого состояния в больших языковых моделях

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Аннотация. Селективные модели пространства состояний, такие как Mamba, стали эффективной альтернативой механизмам внимания в больших языковых моделях, обеспечивая линейную по времени сложность вывода. Однако недавние работы показали, что зависящий от входа механизм дискретизации создаёт ранее не обнаруженную поверхность атаки: короткие враждебные последовательности токенов способны привести параметр шага дискретизации Δ_t к насыщению, вызывая необратимое «отравление скрытого состояния» (HiSPA), которое стирает предыдущий контекст и распространяется на все последующие токены. Существующие средства защиты, включая детекторы образцов активаций и методы прерывания цепей, обеспечивают эмпирическое смягчение, но не дают формальных гарантий. В настоящей работе представлена LipMamba — архитектура селективной SSM с ограничением Липшица, дополненная сертификатами обобщения PAC-Байеса, которые в совокупности впервые обеспечивают гарантированную архитектурную защиту от отравления скрытого состояния. Сделано три вклада. Во-первых, выведены точные оценки Липшица для слоёв селективных SSM, явно учитывающие зависимость от

входа тройки $(\Delta_t, \mathbf{B}_t, \mathbf{C}_t)$. Во-вторых, доказана теорема о сертифицированном иммунитете к отравлению, утверждающая, что при спектрально-ограниченных параметризациях и обрезке шага $\Delta_t \leq \Delta_{\max}$ ни один триггер длиной меньше аналитически вычисленной границы ℓ^* не может снизить норму скрытого состояния ниже выбранного порога безопасности $\alpha_{\min}$. В-третьих, оценки РАС-Байеса в стиле Мак-Аллестера расширены на SSM с ограничением Липшица, давая невырожденные численные значения на масштабе 130–370 миллионов параметров. Оценка на наборах RoBench-25, HarmBench, JailbreakBench и четырёх базах данных систем обнаружения вторжений (CIC-IDS2017, Edge-IoTset, UNSW-NB15, TON_IoT) демонстрирует точ-

ность с сертификатом 80,4% при сертифицированном радиусе $\epsilon^* = 0,18$ для атак HiSPA при длине триггера $\ell = 24$, лишь с накладными расходами в 4,7% по перплексии относительно неограниченной Mamba и со снижением показателя успеха атаки на 61,2 процентных пункта. LipMamba интегрирована в платформу RobustIDPS.ai как сертифицированная замена модели MambaShield для обработки потоков с задержкой 0,6 мс на одно ядро ЦП.

Ключевые слова: селективные модели пространства состояний, Mamba, непрерывность Липшица, РАС-Байес, гарантированная устойчивость, отравление скрытого состояния, большие языковые модели, системы обнаружения вторжений.